

Chapter 17 - POKEY and SAP Playback

POKEY is the Atari 8-bit sound chip - four square-wave channels with seven distortion modes each, a 17-bit polynomial noise generator, and a small but versatile set of clock-routing modes that let two channels combine into a 16-bit oscillator. It is the chip behind the Atari 400/800/XL/XE sound and the SAP music format used to preserve their songs.

17.1 What POKEY can show

Item	Value
Channels	4
Distortion modes	8 per channel (combinations of polynomial taps and pure tone)
Frequency range	8-bit divider per channel
Volume control	Per-channel 4-bit volume
Clock options	64 kHz, 15 kHz, or 1.79 MHz (per channel)
Pair modes	Channels 1+2 and 3+4 can combine to 16-bit
Filter modes	High-pass on channels 1 and 2
Polynomial RNG	Readable through POKEY_RANDOM

17.2 The register block

The chip sits in an eleven-byte block at 0xF0D00–0xF0D0A.

Address	Name	Purpose
0xF0D00	POKEY_AUDF1	Channel 1 frequency divider.
0xF0D01	POKEY_AUDC1	Channel 1 control (distortion + volume).
0xF0D02	POKEY_AUDF2	Channel 2 divider.
0xF0D03	POKEY_AUDC2	Channel 2 control.
0xF0D04	POKEY_AUDF3	Channel 3 divider.
0xF0D05	POKEY_AUDC3	Channel 3 control.
0xF0D06	POKEY_AUDF4	Channel 4 divider.
0xF0D07	POKEY_AUDC4	Channel 4 control.
0xF0D08	POKEY_AUDCTL	Master audio control.
0xF0D09	POKEY_PLUS_CTRL	POKEY+ mode enable (bit 0).
0xF0D0A	POKEY_RANDOM	Read-only polynomial / RNG tap.

17.2.1 AUDC (channel control) bits

Bits	Field	Meaning
0–3	VOLUME	Volume, 0–15.
4	VOLUME_ONLY	Force DC output (no oscillator); the channel just produces the volume value.
5–7	DISTORTION	One of eight distortion modes (table below).

DISTORTION	Algorithm
0	17-bit poly + 5-bit poly
1	5-bit poly only
2	17-bit poly + 4-bit poly (most metallic)
3	5-bit poly + 4-bit poly
4	17-bit poly only (white noise)
5	Pure square wave
6	4-bit poly only (buzzy)
7	17-bit poly + pulse (50% duty)

17.2.2 AUDCTL (master) bits

Bit	Field	Meaning
0	CLOCK_15KHZ	Use 15 kHz base clock instead of 64 kHz.
1	HIPASS_CH1	High-pass filter ch1 by ch3.
2	HIPASS_CH2	High-pass filter ch2 by ch4.
3	CH4_BY_CH3	Ch4 clocked by ch3 (joined as a 16-bit oscillator).
4	CH2_BY_CH1	Ch2 clocked by ch1 (16-bit).
5	CH3_179MHZ	Ch3 uses 1.79 MHz clock.
6	CH1_179MHZ	Ch1 uses 1.79 MHz clock.
7	POLY9	Use 9-bit poly instead of 17-bit.

17.2.3 Clocks

Constant	Value (Hz)
POKEY_CLOCK_NTSC	1,789,773
POKEY_CLOCK_PAL	1,773,447

The output frequency of a channel is

$$f = \text{clock} / ((1 + \text{divider}) \times \text{prescaler})$$

with prescaler chosen by AUDCTL: 28 for 64 kHz mode, 114 for 15 kHz mode, or 1 for the 1.79 MHz modes.

17.3 The POKEY+ extension

Bit 0 of POKEY_PLUS_CTRL enables **POKEY+**: extended volume, DAC mode, and additional distortion combinations. Bit-exact POKEY behaviour resumes when the bit is cleared.

17.4 SAP playback

The SAP (Slight Atari Player) format is a wrapper around an Atari 6502 routine and the POKEY register changes it produces. The chip's player handles loading and per-frame replay:

Address	Name	Purpose
0xF0D10	SAP_PLAY_PTR	Address of the SAP data.
0xF0D14	SAP_PLAY_LEN	Length in bytes.
0xF0D18	SAP_PLAY_CTRL	Bit 0 = start, bit 1 = stop, bit 2 = loop.
0xF0D1C	SAP_PLAY_STATUS	Bit 0 = busy, bit 1 = error.
0xF0D20	SAP_SUBSONG	Subsong number (0–255).

17.5 BASIC keywords

Form	Effect
POKEY ch , $freq$, $ctrl$	Set channel divider and control byte.
POKEY CTRL $value$	Write the master AUDCTL byte.
POKEY PLUS ON / POKEY PLUS OFF	Enable / disable POKEY+.
SAP $addr$ [, len [, $subsong$]]	Start SAP playback.
SAP STOP	Stop SAP playback.

A BASIC fragment that plays a pure tone through channel 1:

```
10 POKE &H000F0800, 1      : REM AUDIO_CTRL = on
20 POKE &H000F0D08, &H00    : REM AUDCTL: 64 kHz, no joins
30 POKEY 0, 121, &HAF       : REM ch 1, divider 121, distortion=5 (pure), vol=15
```

17.6 Putting it together

POKEY's eight distortion modes plus its 16-bit pair modes and 1.79 MHz clock give it more pitch range and timbre variety than any of the other heritage chips. For melodic tracks, set the distortion to 5 (pure tone) on the channels you want clean. For drums and effects, use 4 (white noise) or 2 (metallic noise). SAP playback handles existing Atari music directly.

The next chapter covers AHX, the Amiga chip-tune format.